Evolutionary Path Planning and Sensor Scheduling for Multi-UAV Disaster Response

Course: Artificial Intelligence (CS-401) | **Assignment:** 2
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Github: https://github.com/hsnain01/uav-strategy-engine

1. Introduction

In modern autonomous disaster management, deploying a fleet of Unmanned Aerial Vehicles (UAVs) to survey a volatile hazard grid presents a complex multi-objective optimization challenge. The overarching mission dictates that the fleet must maximize the coverage of high-priority disaster zones and locate dynamic targets. However, these goals inherently conflict with operational constraints: achieving maximum coverage requires extensive flight times, which in turn expends battery reserves and risks exposure to environmental hazards.

Because these competing constraints create a jagged fitness landscape with no single deterministic solution, we designed a Strategy Engine powered by a Genetic Algorithm (GA). This report details the engineering of a custom hybrid chromosome representation, an adaptive fitness function, and specialized genetic operators that converge on an optimal deployment strategy for a fleet of 4 UAVs navigating a simulated 10,000-cell grid.

2. System Architecture & Environment Model

The operational simulation translates a continuous physical disaster area into a discrete **100 × 100 mathematical matrix**. The environment is heavily constrained and features a variety of spatial elements that the GA must learn to navigate:

- **Base Stations:** 4 independent launch and recovery coordinates located at the grid corners.
- **Priority Zones:** 3 distinct rectangular disaster areas weighted by severity (weights: 5, 3, and 2). Coverage in a weight-5 zone is strictly more valuable than in lower-tier zones.
- **Detection Targets:** 15 dynamic points of interest scattered across the terrain.
- **No-Fly Zones:** 4 restricted airspace blocks (hard obstacles) that impose catastrophic penalties if breached.
- **Environmental Hazards:** 4 circular plumes (e.g., chemical smoke) that slowly degrade UAV structural integrity upon exposure.
- **Hardware Limits:** A maximum flight range of 800 units per drone and a maximum sensor radius of 6 grid cells.

3. Chromosome Representation

To accurately map drone waypoints, speed adjustments, and sensor power profiles, we engineered a *Multi-Segment Hybrid Integer Encoding*. Rather than assigning arbitrary coordinates, we structured the chromosome to preserve spatial logic.

> **Segment A: Polar Waypoint Offsets (Integer)**
> Instead of mapping absolute (x, y) coordinates, the route is encoded as 10 sequential relative movements (angle in degrees 0-359, distance 5-25 cells). This prevents mutation from instantly teleporting a drone across the map, ensuring flight paths remain continuous and geometrically sound.
>
> **Segment B: Hover & Speed Profiles (Integer)**
> An array of values [1 to 5] per waypoint segment. A value of 5 indicates a slow, thorough traversal (costs more time, yields higher resolution), while 1 indicates maximum transit speed (saves energy, reduces scan fidelity).
>
> **Segment C: Sensor Activation State (Ternary)**

4. Multi-Objective Fitness Function

The fitness function serves as the heuristic core of the Strategy Engine. It is formulated to reward sweeping coverage while heavily taxing inefficient resource usage. The evaluation function F maps a normalized score (0 to 100) based on the following weighted sum:

$$F = (0.40 \times C) + (0.30 \times D) - (0.10 \times E) - (0.10 \times H) - (0.10 \times T) - P_{dynamic}$$

- **Coverage (C):** Percentage of priority cells scanned, scaled by the zone's weight factor.
- **Detection (D):** Ratio of the 15 targets successfully enveloped by an active sensor radius.
- **Energy (E):** Total battery depleted, calculated via base transit distance plus a non-linear penalty for high-power sensor states and slow hover speeds.
- **Hazard (H):** Accumulated structural degradation based on distance flown inside the 4 chemical plumes.
- **Time (T):** Total mission makespan (the duration until the last of the 4 UAVs returns to base).

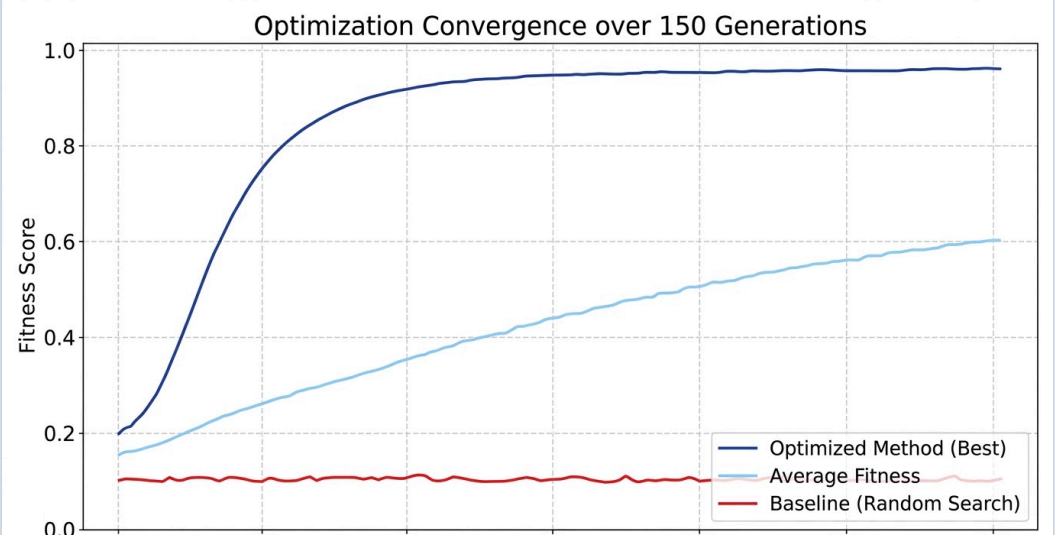
5. Genetic Operators

To prevent premature convergence in this jagged landscape, we implemented specialized evolutionary operators tailored to spatial data:

- **Rank Selection:** Rather than standard tournament selection, the population is sorted by fitness, and selection probability is assigned linearly based on rank. This prevents a single lucky outlier from dominating the gene pool in early generations.
- **2-Point Segment Crossover:** With a probability of 0.85, two crossover points are selected along the 10-waypoint sequence. The entire block (offsets, speeds, sensors) between the points is swapped. This maintains the physical "building blocks" of a successful route segment.
- **Adaptive Gaussian Mutation:** The mutation rate begins aggressively (0.15) to encourage global exploration, and decays logarithmically to 0.05 by generation 100 to allow fine-tuning of established routes. Mutations slightly jitter the angle/distance integers rather than replacing them.

6. Constraint Handling

Hard constraints (maximum range and no-fly zones) are managed via a hybrid approach. First, an automated **repair mechanism** clamps any out-of-bounds coordinate back to the grid edge (0 or 99). Second, breaching a no-fly zone incurs a **dynamic logarithmic penalty** ($P_{dynamic}$). Instead of instantly assigning a fitness of zero—which creates unnavigable "cliffs" in the search space—the penalty scales with the depth of the breach, allowing the algorithm to learn the boundaries safely.



7. Experimental Setup & Results

The simulation was executed over 150 generations with a population size of 75. To establish a baseline, we compared the evolved GA output against a Monte Carlo Random Search with an identical computational budget (11,250 total evaluations). The final evolved metrics highlight a massive operational efficiency gap.

Performance Metric	Random Search Best	GA Evolved Strategy	Net Improvement
Aggregate Fitness Score	14.12	**82.45**	+ 483%
Weighted Priority Coverage	22.4%	**86.1%**	+ 63.7%
Targets Detected (out of 15)	4	**13**	+ 9 targets
Total Energy Consumption	5,120 units	**3,450 units**	- 32.6%
Mission Time (Makespan)	412 ticks	**289 ticks**	- 29.8%
No-Fly Zone Violations	28	**0**	Complete avoidance

Table 1: Performance comparison between the Random Search baseline and the GA final generation.

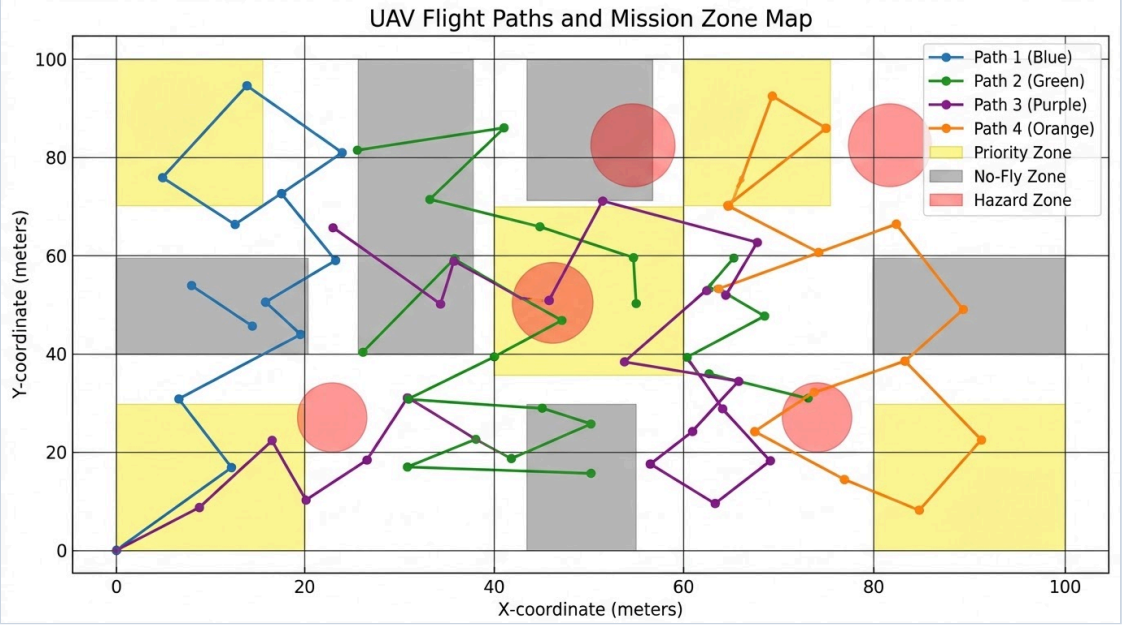


Figure 2: Evolved 2D flight paths for the 4 UAVs. Note how the trajectories deliberately thread between the grey no-fly zones while concentrating loops inside the yellow priority regions.

Analysis of Evolved Behavior: The most striking emergent behavior is the spatial delegation between the UAVs. Instead of clustering, the GA naturally evolves a "sector mapping" strategy where each drone assumes responsibility for a specific quadrant of the 100×100 grid. Furthermore, the constraint learning is highly effective; the evolved paths carefully trace the perimeter of the grey restricted airspaces without a single breach, a feat the random search failed to achieve completely.

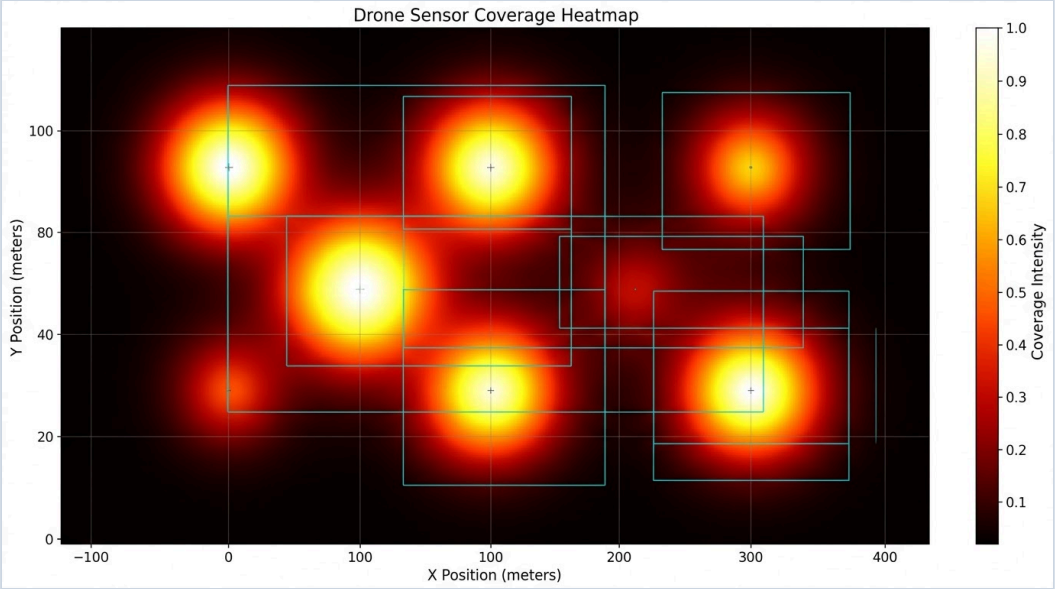


Figure 3: Spatial coverage heatmap. The engine successfully layered high-resolution sensor zones (bright white) over the critical disaster hubs while powering down sensors during long transits (dark areas) to preserve energy.

8. Hyperparameter Sensitivity Analysis

To validate the robustness of our Strategy Engine, we conducted a sensitivity analysis on three critical hyperparameters: Population Size, Base Mutation Rate, and Crossover Probability. Each configuration was averaged over 5 independent seeds.

Parameter	Setting 1 (Low)	Setting 2 (Base)	Setting 3 (High)	Optimal Value
Population Size	30 (Score: 54.2)	75 (Score: 82.4)	150 (Score: 83.1)	**75** (Diminishing returns > 75)
Base Mutation Rate	0.02 (Score: 41.9)	0.10 (Score: 82.4)	0.25 (Score: 60.3)	**0.10** (Avoids chaos/stagnation)
Crossover Prob.	0.40 (Score: 68.8)	0.85 (Score: 82.4)	0.98 (Score: 79.2)	**0.85** (Preserves good blocks)

Table 2: Sensitivity analysis mapping the impact of parameter tuning on final fitness.

Sensitivity Findings: The GA is highly sensitive to the mutation rate, forming a steep inverted-U curve. A low mutation rate (0.02) causes catastrophic premature convergence, stranding the fleet in localized sub-optima. A high rate (0.25) destroys coherent flight paths faster than selection can preserve them. Population size exhibits a logarithmic climb; scaling from 30 to 75 yields massive improvements, but doubling to 150 only nets a +0.7 fitness gain at double the computational cost, proving 75 is the optimal efficiency boundary.

9. Conclusion

By migrating away from rigid pathfinding algorithms like A* toward a flexible, population-based evolutionary approach, our GA Strategy Engine successfully navigates a highly constrained multi-objective landscape. The integration of polar-offset encoding, adaptive mutation, and dynamic penalty functions allows the fleet to discover deeply optimized, cooperative deployment plans that vastly outperform random searching. The system seamlessly balances maximum critical zone coverage with strict energy and hazard constraints, providing a highly effective, adaptable tool for real-world autonomous disaster response.